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POWER SYSTEM-1 LAB

EEE-308

Power factor correction Using Power World
Simulator.

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I. **Prelude:** In electrical power systems, the power factor is a crucial quantity that indicates how effectively the linked load uses electrical power. It is an infinite amount. The power factor of a system is contingent upon the kind of connected load, whether resistive, inductive, or capacitive.

II. **Objective:** Power factor correction Using Power World Simulator

III. Introduction

The power factor is a measure of energy efficiency, according to theory. Typically, it is given as a percentage, and the lower the percentage, the less efficiently energy is used. Formally, power factor (PF) is the ratio of working power in kilowatts (kW) to perceived power in kilovolt amperes (kVA). Apparent power, often known as demand, is the quantity of energy utilized to operate machinery and equipment during a certain time. It is determined through multiplication ($kVA = V \times A$). The outcome is reported in kVA units. PF is the ratio between the actual power consumed in a circuit and the perceived power given to the circuit. The design of any power transmission system is highly dependent on the size of the line current as defined by the applied loads. Due to the resistance of the transmission lines, greater currents result in increased power losses (by a squared factor, since $P = I^2R$) in the transmission lines. Additionally, heavier currents need bigger conductors, which increases the quantity of copper required for the system, as well as an increase in the utility company's producing capacity.

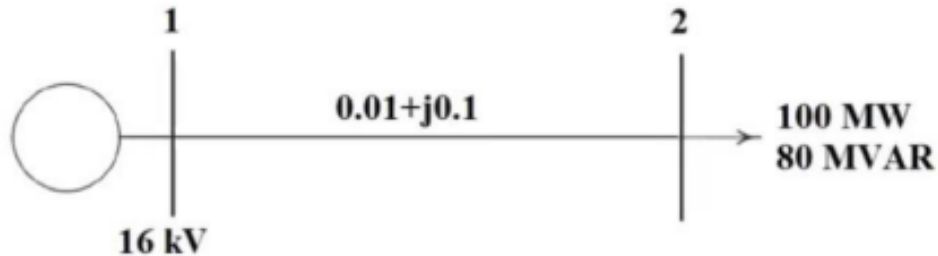
a.) Leading Power Factor: A leading power factor indicates that the circuit's load is capacitive in nature. As the load provides reactive power to the circuit, the reactive component (Q) of Electrical power will be negative in this instance. If there is more capacitive reactance than inductive reactance in an electrical circuit, then the circuit's power factor will be leading. In this situation, the running current will lead the voltage phasor by an angle, while in a pure capacitive circuit, the operating current would lead the voltage by an angle of $= 90$ degrees.

Idempotent matrix characteristics

b.) Lagging Power Factor: A lagging power factor indicates an inductive load in the circuit. As the load will use reactive power, the reactive component (Q) of Electrical power will be positive in this instance. If the electrical circuit has more inductive reactance than capacitive reactance, then the power factor of the circuit will be trailing. In this circumstance, the operational current will trail the voltage phasor by an angle of. In pure inductive circuits, the current trails the voltage by 90 degrees.

c.) Unity Power Factor: A power factor of unity indicates that the load in the circuit is completely resistive. As the load will utilize active power, the reactive component (Q) of Electrical power will be 0 in this instance. In a circuit where the power factor is unity, the complete apparent power will be consumed. In an electrical circuit, if the inductive reactance and capacitive reactance are equal, the circuit will be in resonance and regarded totally resistive, resulting in a power factor of unity. The operational current will be in phase with the voltage phasor in this instance. In this instance, will equal 0 degrees.

- IV. Practice Problem 1:** A single-phase source delivers 100 MW to a load operating at a power factor of 0.75 lagging. Calculate the reactive power to be delivered by a capacitor connected in parallel with the load (70 MW) in order to raise the source power factor to 0.95 lagging. Assume that the source voltage is constant, and the line impedance between the source and load as mentioned below.



Solution to Practice Problem 1:

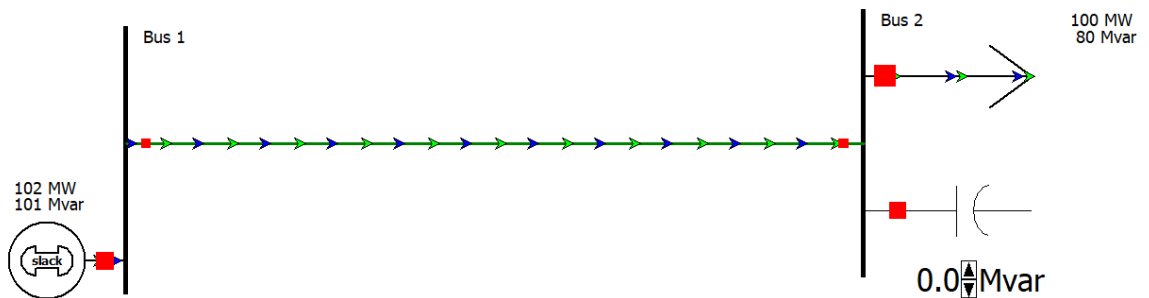


Fig: 1 (a) Power System with no capacitor therefore the generator delivers all reactive power

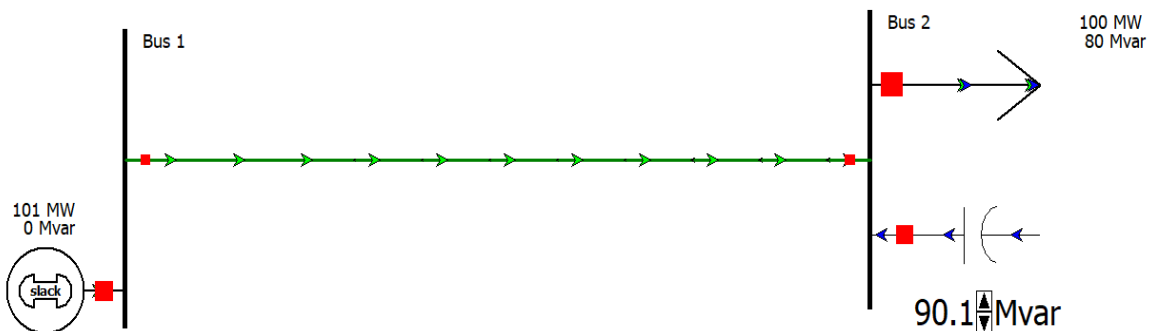


Fig: 1 (b) Power System with capacitor it delivers all the reactive power

The capacitor delivers 90.1 MVAR reactive power to the power system.

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- V. Practice Problem 2:** A $20+j10$ MVA load is supplied at 20 kV through a feeder as shown in Figure. The load is compensated with a capacitor whose output, can be varied in 1 MVAR steps between 0 and 20.0 MVARs.

Determine the effect of adding capacitor bank?

Solution to Practice Problem 2:

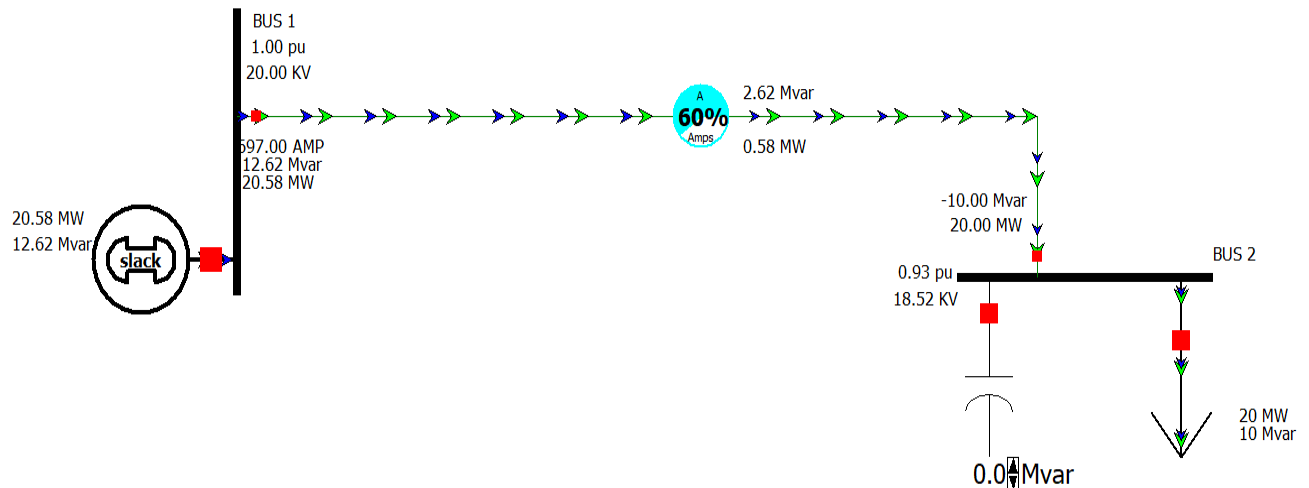


Fig: 2 (a) Power System with no capacitor therefore the generator delivers all reactive power

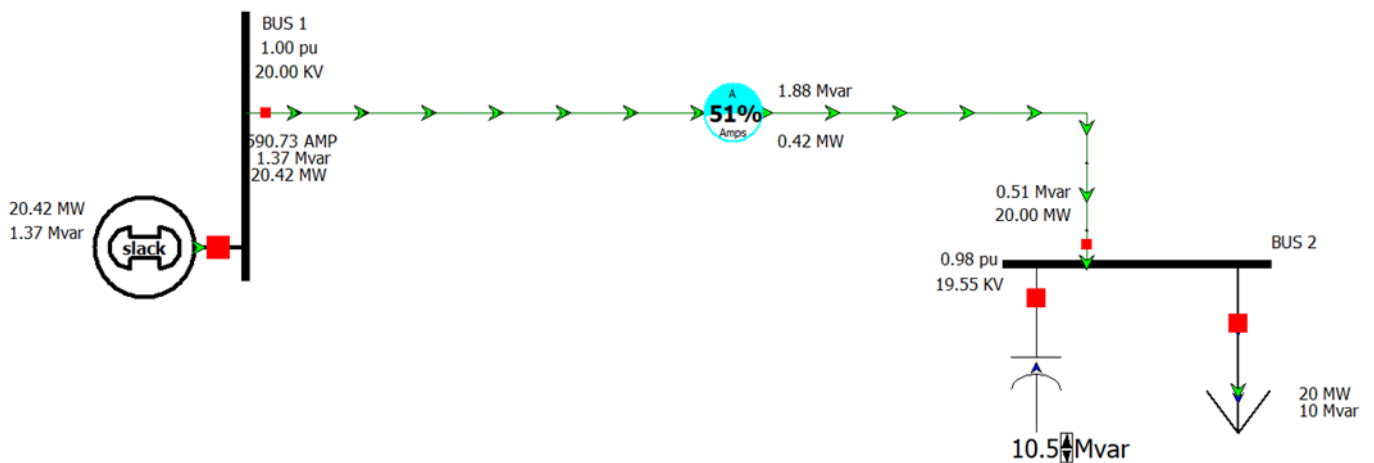


Fig: 2 (b) Power System with capacitor it delivers all the reactive power

The capacitor delivers 10.5 MVAR reactive power to the power system.

VI. Discussion:

Since most industrial and household loads are inductive the addition of capacitor connected parallelly to the load to improve the power factor. In practice problem-1 and 2, the load requires 80 MVAR and 10.5 MVAR respectively in problem 1 and 2 are the reactive loads, without the capacitor bank the reactive power was used to be delivered by the generator. The reactive load supplied by the generator will drop the Power Factor greatly. When the capacitor is added the reactive power is delivered by the capacitor